

we refer to β exposure – meaning all tasks directly exposed via tools like ChatGPT or the OpenAI Playground are considered twice as exposed as tasks requiring some complementary innovation.

3.4 Limitations of our methodology

3.4.1 Subjective human judgments

A fundamental limitation of our approach lies in the subjectivity of the labeling. In our study, we employ annotators who are familiar with LLM capabilities. However, this group is not occupationally diverse, potentially leading to biased judgments regarding LLMs’ reliability and effectiveness in performing tasks within unfamiliar occupations. We acknowledge that obtaining high-quality labels for each task in an occupation requires workers engaged in those occupations or, at a minimum, possessing in-depth knowledge of the diverse tasks within those occupations. This represents an important area for future work in validating these results.

3.4.2 Measuring LLMs with GPT-4

Recent research indicates that GPT-4 serves as an effective discriminator, capable of applying intricate taxonomies and responding to changes in wording and emphasis (OpenAI, 2023b). The outcomes of GPT-4 task classification are sensitive to alterations in the rubric’s wording, the prompt’s order and composition, the presence or absence of specific examples in the rubric, the level of detail provided, and the definitions given for key terms. Iterating on the prompt, based on observed outcomes in a small validation set, can enhance the agreement between model outputs and the rubric’s intent. Consequently, there are slight differences between the rubric presented to humans and the one used for GPT-4. This decision was made deliberately to guide the model towards reasonable labels without excessively influencing human annotators. As a result, we use multiple annotation sources, but none should be considered the definitive ground truth relative to the others. In this analysis, we present results from human annotators as our primary results. Further improvement and innovation in crafting effective rubrics for LLM classification remains possible. Still, we observe a high degree of agreement between human ratings and GPT-4 ratings at the occupation level concerning overall exposure to LLM systems (see Table 2, Figure 2).

3.4.3 Additional Weaknesses

- **Validity of task-based framework.** It is unclear to what extent occupations can be entirely broken down into tasks, and whether this approach systematically omits certain categories of skills or tasks that are tacitly required for competent performance of a job. Additionally, tasks can be composed of sub-tasks, some of which are more automatable than others. Some tasks may function as pre-cursor to other tasks, such that the completion of downstream tasks is dependent on precursor tasks. If indeed, the task-based breakdown is not a valid representation of how most work in an occupation is performed, our exposure analysis would largely be invalidated.
- **Lack of expertise and task interpretation.** Human annotators were mostly unaware of the specific occupations mapped to each DWA during the labeling process. This led to unclear logic for aggregating tasks and occupations, as well as some evident discrepancies in labels, demonstrated in Table 1. We experimented with various aggregation methods and discovered that even with a maximum-matching approach (taking the matching human<>model label if one existed), the agreement remained relatively consistent. Ultimately, we collected additional labels for task/occupation pairs where there was significant disagreement.

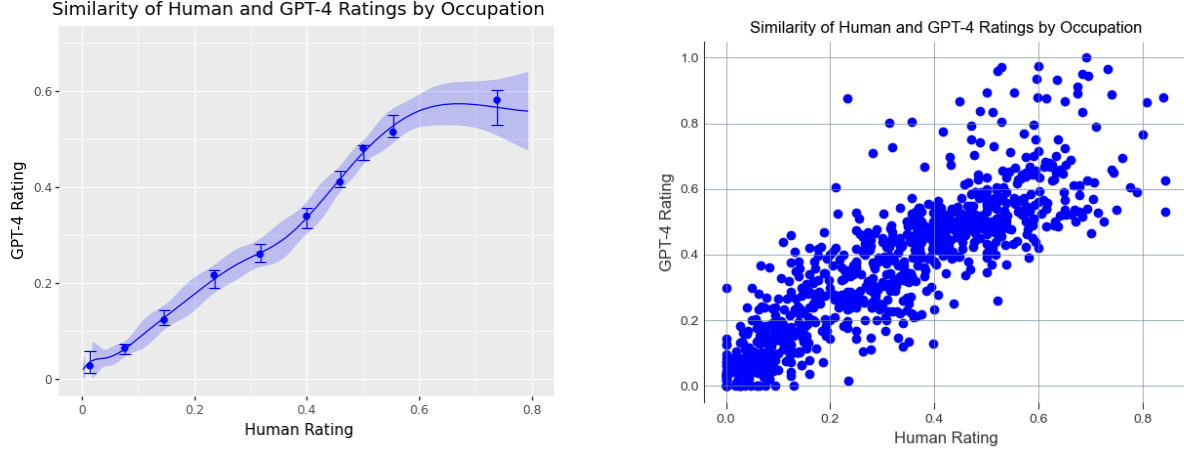


Figure 2: Human raters (x-axis) and GPT-4 ratings (y-axis) show a high degree of agreement about LLM exposure by occupation. We compute occupation-level exposure in these figures by averaging the task-level exposures under the β method. O*NET designates some tasks as "core" and others "supplemental". Core tasks are given twice the weight of supplemental tasks, and all weights sum to one. Near the highest levels of exposure following the β method of aggregating exposure scores to occupations, GPT-4 ratings tend to be lower than Human ratings. We present the raw scatter plot and the binscatter. Near the top end of exposure ratings, humans are on average more likely to rate an occupation as exposed.

- **Forward-looking and subject to change, with some early evidence.** Accurately predicting future LLM applications remains a significant challenge, even for experts (OpenAI, 2023b). The discovery of new emergent capabilities, changes in human perception biases, and shifts in technological development can all affect the accuracy and reliability of predictions regarding the potential impact of LLMs on worker tasks and the development of LLM-powered software. Our projections are inherently forward-looking and based on current trends, evidence, and perceptions of technological possibilities. As a result, they may change as new advancements arise in the field. For example, some tasks that seem unlikely for LLMs or LLM-powered software to impact today might change with the introduction of new model capabilities. Conversely, tasks that appear exposed might face unforeseen challenges limiting language model applications.
- **Sources of disagreement.** While we did not rigorously examine sources of disagreement, we found a few places where humans and the model tended to get "stuck" in their assessments:
 - Tasks or activities where while an LLM could theoretically help or accomplish the task, adopting it to do so would require multiple people to change their habits or expectations (e.g. meetings, negotiations),
 - Tasks or activities where there is currently some regulation or norm that requires or suggests human oversight, judgment or empathy (e.g. making decisions, counseling), and
 - Tasks or activities where there already exists a technology that can reasonably automate the task (e.g. making reservations).

4 Results

General-purpose technologies are relatively rare and characterized by their pervasiveness, improvement over time, and the development of significant co-invention and spillovers (Lipsey et al., 2005). Our assessment of

LLMs' potential impact on the labor market is limited since it does not consider total factor productivity or capital input potential. In addition to their influence on labor, LLMs may also influence these dimensions.

At this stage, some general-purpose technology criteria are easier to evaluate than others. Our primary focus at this early stage is to test the hypothesis that LLMs have a pervasive influence on the economy, similar to the approach taken by (Goldfarb et al., 2023), who analyzed machine learning diffusion through job postings to assess its status as a general-purpose technology. Instead of using job postings or studying machine learning in general, we employ the task evaluation approach with both human and GPT-4 annotations. This analysis may reveal whether the impacts are limited to a specific set of similar tasks or occupations or if they will be more widespread.

Our findings suggest that, based on their task-level capabilities, LLMs have the potential to significantly affect a diverse range of occupations within the U.S. economy, demonstrating a key attribute of general-purpose technologies. In the following sections, we discuss results across various roles and wage structures. Additional results on the relative exposure of industries within the U.S. economy can be found in Appendix C.

4.1 Summary Statistics

Summary statistics for these measures can be found in Table 3. Both human and GPT-4 annotations indicate that average occupation-level α values fall between 0.14 and 0.15, suggesting that, on average, approximately 15% of tasks within an occupation are directly exposed to LLMs.⁶ This figure increases to over 30% for β and surpasses 50% for ζ . Coincidentally, human and GPT-4 annotations also tag between 15% and 14% of total tasks in the dataset as being exposed to LLMs. Based on the β values, we estimate that 80% of workers belong to an occupation with at least 10% of its tasks exposed to LLMs, while 19% of workers are in an occupation where over half of its tasks are labeled as exposed.

We ran one set of analyses using O*NET's "Importance" scores but did not find significant changes to our findings. Though we do acknowledge that not weighting relative importance of a task to a given occupation yields some curious results (e.g. ranking Barbers as having reasonably high exposure).

Although the potential for tasks to be affected is vast, LLMs and LLM-powered software must be incorporated into broader systems to fully realize this potential. As is common with general-purpose technologies, co-invention barriers may initially impede the rapid diffusion of GPTs into economic applications. Furthermore, predicting the need for human oversight is challenging, especially for tasks where model capabilities equal or surpass human levels. While the requirement for human supervision may initially slow down the speed at which these systems diffuse through the economy, users of LLMs and LLM-powered systems are likely to become increasingly acquainted with the technology over time, particularly in terms of understanding when and how to trust its outputs.

4.2 Wages and Employment

In Figure 3, we present exposure intensity across the economy. Each point on the plot displays occupational exposure, where the point's x-axis value represents the share of an occupation's tasks that are exposed (at each level α , β , and ζ) and the point's y-axis value represents the share of all US occupations with that share of tasks exposed. For example, human annotators determined that 2.3% of occupations are α_{50} -exposed, 21.6% are β_{50} -exposed, and 47.3% are ζ_{50} -exposed, where the threshold of 50% comes from the x-axis and the percentage of occupations comes from the y axis. At any given point on the x-axis, the vertical distance between the α and the ζ represents the exposure potential attributable to tools and applications beyond direct exposure to LLMs. All tasks within an occupation in this figure are given equal weight.

⁶We compute occupation-level scores for Table 3 assigning double the weight to tasks designated as "core" by O*NET as tasks designated "supplemental". All tasks weights sum to one within an occupation.

Occupation Level Exposure				
	Human		GPT-4	
	mean	std	mean	std
α	0.14	0.14	0.14	0.16
β	0.30	0.21	0.34	0.22
ζ	0.46	0.30	0.55	0.34

Task Level Exposure				
	Human		GPT-4	
	mean	std	mean	std
α	0.15	0.36	0.14	0.35
β	0.31	0.37	0.35	0.35
ζ	0.47	0.50	0.56	0.50

Table 3: Summary statistics of our human and model exposure data. Tasks designated as core tasks for an occupation are given twice the weight as those indicated to be supplemental in the O*NET task file.

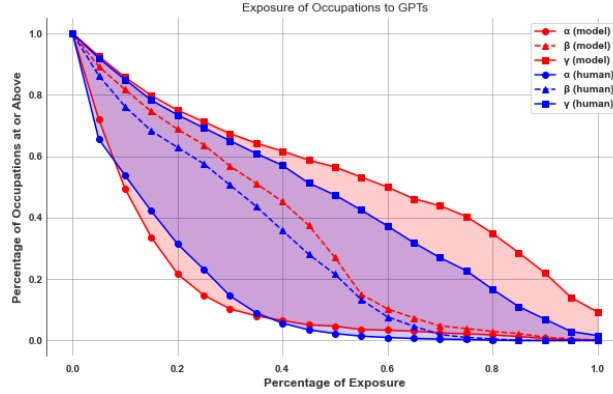


Figure 3: Exposure intensity across the economy, displayed in terms of percent of affected occupations. A given data point gives the percent of occupations with exposure below the given threshold. A previous version of this paper had two labels reversed in the chart, flipping human and model responses. In this figure, all tasks within an occupation are given equal weight.

Aggregated at the occupation level, human and GPT-4 annotations exhibit qualitative similarities and tend to correlate, as demonstrated in Figure 4. Human annotations estimate marginally lower exposure for high-wage occupations compared to GPT-4 annotations. While there are numerous low-wage occupations with high exposure and high-wage occupations with low exposure, the overall trend in the binscatter plot reveals that higher wages are associated with increased exposure to LLMs.⁷

The potential exposure to LLMs seems to have little correlation with current employment levels. In Figure 4, both human and GPT-4 ratings of overall exposure are aggregated to the occupation-level (y-axis) and compared with the log of total employment (x-axis). Neither plot reveals significant differences in LLM exposure across varying employment levels.

⁷In aggregating tasks to the occupation-level, we assign half the weight to O*NET supplemental tasks as we do for core tasks.